

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Spring 2025
Course Code: CSE-4745 Course Title: Numerical Methods
Total marks: 50 Time: 2 hours 30 minutes

[Answer all question from Group-A and Group-B; Separate answer script must be used for Group-A and Group-B.]

Group-A

- | | | |
|--|-------|------|
| | Marks | CLOs |
| 1.a) State the condition for a matrix to be invertible.
When is a system said to have:
(a) Unique solution
(b) No solution
(c) Infinite solutions | 3 | CO1 |
| b) Compare and contrast the Gauss elimination method with the Gauss-Jordan method | 3 | CO2 |
| c) Solve the following system of equation by using Matrix Inversion Method
$3x_1 + 6x_2 + x_3 = 16$
$2x_1 + 4x_2 + 3x_3 = 13$
$x_1 + 3x_2 + 2x_3 = 9$ | 4 | CO2 |

OR

- | | | |
|--|---|-----|
| c) Solve the following system of equation by using Basic Gauss elimination with partial pivoting
$2x_1 + 3x_2 + 4x_3 = 5$
$3x_1 + 4x_2 + 5x_3 = 6$
$4x_1 + 5x_2 + 6x_3 = 7$ | 4 | CO2 |
| 2.a) Compare curve fitting with interpolation. | 2 | CO1 |
| b) Describe the <i>least square method</i> to fit a straight line. | 4 | CO1 |

OR

- | | | |
|---|---|-----|
| b) Describe the <i>least square method</i> to fit a parabola. | 4 | CO1 |
| c) The following data shows battery health.(y) after months of usage (x): | 4 | CO2 |

x (Months)	0	1	2	3	4
y (%)	100	92	83	75	60

Fit a straight line $y = a + bx$ using the least squares method.

Group-B

- | | | |
|---|---|-----|
| 3.a) Determine the <i>first order</i> and <i>second order derivatives</i> using Newton's forward interpolation formula. | 4 | CO3 |
|---|---|-----|

A ball is thrown and its height is recorded at different times.

4 CO4

b)

Time (s)	0	1	2	3	4
Height (m)	0	20	35	30	15

Find *velocity* and *acceleration* at $t = 2$ seconds.

- c) Describe how numerical differentiation helps in locating the local *minima* or *maxima* of a tabulated function. 2 CO3

- 4.a) Why does Simpson's 1/3 rule require an even number of intervals?

- b) Derive the Simpson's 3/8 rule from general quadrature formula

- c) Use **Simpson's 3/8 rule** to evaluate the integral $\int_0^1 \frac{1}{1+x^2}$ correct up to 3 significant figures taking 3 sub intervals by trapezoidal

Handwritten calculations for Simpson's 3/8 rule:

$$\begin{array}{r} 0 \\ 20 \\ 35 \\ 15 \end{array} \quad \begin{array}{r} 20 \\ 15 \\ 5 \\ -15 \end{array} \quad \begin{array}{r} -5 \\ -2 \\ 10 \\ -15 \end{array} \quad \begin{array}{r} 15 \\ 1 \\ 4 \\ 5 \end{array}$$

- 5.a) Describe how Euler's method approximates the solution of an ODE. What is its main limitation? 2 CO3

OR

- a) Explain why the Taylor's series method, though conceptually important, is rarely used in practice for solving ODEs numerically. 2 CO3

- b) Describe Euler's method for the numerical solution of ordinary differential equations with an example. 4 CO3

- c) Given $\frac{dy}{dx} = x + y + xy$, where $y(0) = 1$, $h = 0.5$, find $y(1)$ by Heun's method up to two decimal places. 4 CO4

Handwritten calculations for Heun's method:

$$\begin{array}{r} 0 \\ 0.2 \\ 0.3 \end{array} \quad \begin{array}{r} -15 \\ 12-12+2 \\ 3 \end{array}$$

International Islamic University Chittagong
Center for General Education (CGED)

Semester End Examination: Spring 2025

Course Code: URIH-4701

Time: 2 hours and 30 minutes.

Program: Undergraduate

Course Title: A Survey of Islamic History & Culture

Full Marks: 50

Instructions:

- i. All Questions are Compulsory.
- ii. Figures in the right margin indicate full marks.
- iii. Course Learning Outcome (CLO) and Bloom's levels are mentioned in additional columns.

Bloom's Levels of the Questions.						
Letter of Symbol	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

	Text of the Questions	Marks	Bloom's Level	CLO
1	Assess the administrative reforms of Abdul Malik bin Marwan. How did these reforms contribute to the development of Islamic governance and modern civilization?	10	U	CLO2
2	Analyze the policies of Umar bin Abdul Aziz. How did the policies reflect the legacy of <i>Khulafa-i-Rashedeen</i> ? Or Investigate the causes of the decline and fall of the Umayyad <i>Khilafah</i> .	10	An	CLO3
3	Examine the contributions of Abu Ja'far Al-Mansur as the real founder of the Abbasid <i>Khilafah</i> and his role in the development of Islamic civilization.	10	An	CLO3
4	Evaluate the contributions of Muslim scholars to the development of geography and mathematics. How has modern civilization benefited from these scientific contributions?	10	E	CLO3
5	Write short notes on any two: a) Conquest of India during <i>Khalifah</i> Al-Walid b) Foreign Policy of <i>Khalifah</i> Harun al-Rashid c) <i>Bait-ul-Hikmah</i> .	5 5	E	CLO3

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE, Final Exam, Spring 2025

Course Code: CSE 3633

Course Title: Computer Networks

Time: 2 hours 30 minutes

Full Marks: 50

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Part A

[Answer the questions from the followings]

- | | | | | |
|-------|---|-----|-----|---|
| 1. a) | Explain Address Resolution Protocol with figure | CO1 | U | 5 |
| b) | Analyze traffic regulation with Leaky bucket method. | CO5 | App | 5 |
| 2. a) | What are the key congestion control techniques and mechanisms that can either prevent congestion before it occurs or remove it after it has happened? Explain with examples. | CO2 | An | 5 |
| 2. b) | Write down the advantages and disadvantages of hierarchical routing. Support your answer with an appropriate network diagram to illustrate how hierarchical routing works in a multi-level network. | CO2 | An | 5 |

OR

- | | | | | |
|------|--|-----|----|---|
| 2 a) | Explain the concepts of Anycast, Broadcast, and Multicast routing. Compare their characteristics. Illustrate your answer with suitable examples or diagrams. | CO2 | U | 5 |
| 2 b) | Computer A has 19.5 MB to send on a network and transmits the data in a burst at 6 Mbps. The maximum transmission rate across routers in the network is 4 Mbps. If Computer A's transmission is shaped using a leaky bucket, how much capacity must the queue in the bucket hold to avoid discarding any data? | CO2 | Ev | 5 |

Part B [Answer the questions from the followings]

- | | | | | |
|-------|---|-----|----|---|
| 3. a) | Discuss any three fields of the TCP header. | CO2 | An | 5 |
| 3. b) | Explain TCP connection establishment with three way handshake. | CO4 | An | 5 |
| 4. a) | Explain slow start and congestion avoidance in the context of TCP congestion handling. | CO3 | An | 5 |
| 4. b) | Illustrate how IP packet is forwarded through an MPLS network. | CO5 | An | 5 |
| 5. a) | What are domain resource records? Provide an example of a DNS database of your choice. | CO3 | Un | 5 |
| 5. b) | Discuss the advantages and challenges of using peer-to-peer architecture and Client server architecture within the application layer. | CO4 | Un | 5 |

OR

- | | | | | |
|-------|--|-----|----|---|
| 5. a) | Compare and contrast DNS iterative queries and recursive queries, providing proper examples. | CO3 | Ap | 5 |
| 5. b) | Describe the architecture of email and compare between POP3 and IMAP protocol. | CO4 | Un | 5 |

International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Final Examination, Spring 2025
Course Code: CSE 4747

Course Title: Mathematical Analysis for Computer Science

Total marks: 50

Time: 2 hours 30 minutes

The figures in the right hand margin indicate full marks.

Course Outcomes and Bloom's Levels are mentioned in additional Column

Course Outcomes (COs) of the Questions	
CO1	Solve recurrence relations to estimate the asymptotic bounds of algorithms.
CO2	Apply counting principles to solve problems related to computing.
CO3	Modeling system behavior and Performance analysis of system.

Bloom's Levels of the Questions						
Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analysis	Evaluate	Create

GROUP A

CLO DL MARKS

[Answer the following questions]

1. a) In a best-of-three sports tournament, the IIUC team plays against their rival. The probability that the IIUC team wins the first game is 40%. For the second and third games, the probability depends on the previous game's outcome:
- If IIUC team lost the previous game, they analyze their mistakes and win the next game with probability 70%.
 - If IIUC team won the previous game, they become overconfident and win the next game with probability 50%.
- i) What is the probability that IIUC teams wins the tournament.
ii) What is the probability that the IIUC team wins the tournament, given that they lost the first game?
iii) What is the probability that the tournament is decided in the first two games?
1. b) A miner trapped in a underground mine and finds four tunnels leading from his cell:
- Tunnel A leads to a dead end after 30 feet.
 - Tunnel B leads to a freedom after 40 feet.
 - Tunnel C leads to a dead end after 50 feet.
 - Tunnel D leads to freedom after 120 feet.

Each time the miner picks one tunnel uniformly at random, crawls along it, and if he finds a dead end, he must crawl back to the cell and try again the next day. Find the expected total distance the miner crawls until he reaches freedom.

OR

1. b) Find the probability that in a family of 4 children there will be

- (a) at least 1 boy and
(b) at least 1 boy and 1 girl.

Assume that the probability of a male birth is 0.5

2. a) Each evening, the punctuality of a IIUC campus bus is recorded. If the bus arrives on time today, the probability that it will be late tomorrow is 0.1. If the bus arrives late today, the probability that it will be late tomorrow is 0.6. CO3 Ap 1 + 4

- i) The bus arrived late today. What is the probability that the bus will be late two days from now (the day after tomorrow)?
ii) Compute the steady-state probability vector for this Markov chain. Based on this, comment on the bus's long-term punctuality: what is the probability that the bus is generally late or on time?

2. b) At a company picnic, five sealed lunch bags are placed on a table, but only one has a special golden ticket for a grand prize. An employee picks one bag at random. Before they open it, the event organizer — who knows which bag holds the ticket — removes and opens two of the other bags, showing they are empty. The employee can now either keep their original bag or switch to one of the two unopened bags left on the table. Should the employee stick with their first choice or switch to have a better chance at winning the grand prize? Analyze and explain which option is better. CO2 An 5

OR

2. b) A computer program crashes at the end of each hour of use with probability p , if it has not crashed 4 already. What is the probability that there will be no crash in a given day (assuming, the day starts with the program remaining operational)? In addition, what is the expected time until the program crashes? CO2 An 5

GROUP B

[Answer the following questions]

3. a) Explain the Bayes Theorem. CO2 R 2
3. b) Suppose you roll three fair, independent six-sided dice and consider the following three events: CO3 Ap 4

- B_1 : The first die shows the same number as the second die.
- B_2 : The second die shows the same number as the third die.
- B_3 : The third die shows the same number as the first die.

Show that these events are pairwise independent, but not mutually independent.

3. c) A university's admission data for two departments shows the following: CO2 An 4

Department	Male Applicants	Male Admitted	Female Applicants	Female Admitted
Computer Science	80	56	20	16
Design	20	12	80	24

- i) Calculate the admission rate for males and females within each department

- ii) Calculate the overall admission rate for males and females.
 iii) Explain why this is an example of Simpson's Paradox.

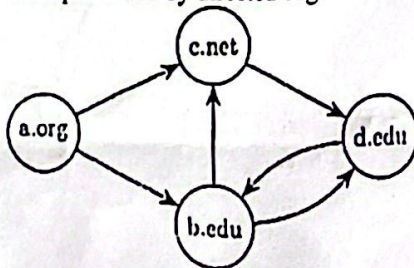
4. a) State the gambler's ruin problem. Formulate a mathematical model to estimate the probability of avoiding ruins. CO3 Ap 5

OR

4. a) Three colleagues — Meher, Taher, and Karim — play a prediction game at lunch. CO3 Ap 5
 Each puts \$3 in a pot, making a total of \$9. They then spin a wheel that lands randomly on red or black. Before the spin, each secretly writes down "red" or "black" to predict the result. The pot is divided equally among those who guessed correctly. If nobody guesses correctly, everyone gets their \$3 back.

Suppose Meher and Taher secretly agree to always pick opposite colors. Explain how this agreement affects the possible outcomes and what this means for Karim's chance of winning compared to theirs.

- 4 b) Formulate a procedure for robust calculation of the ranks of the pages in a Web-graph based on a random walk on that graph. CO3 U 5
 Illustrate how it is meant to work with the following Webgraph where (hyper)links are represented by directed edges.



Make reasonable assumptions as necessary.

OR

- 4 b) Random walks are widely used to estimate the relative importance (or rank) of web pages in a network, as in Google's PageRank algorithm. The basic idea is that a "random surfer" moves from page to page by randomly following outgoing links. CO3 U 5

Consider the following simple Web-graph:

- Page A links to Page B and Page C
- Page B links to Page A
- Page C links to Page B

Represent this Web-graph with a diagram showing pages as nodes and hyperlinks as directed edges. Now, answer the following questions:

- i) Explain how a random walk can be used on this graph to estimate the rank of each page.

Describe how the probability of being on each page stabilizes over many steps.

5. a) Define Markov chain and Markov property. Assume that today is sunny, then there is 90% chance it will sunny tomorrow and 10% chance of rainy. if it doesn't sunny today, then there is a 50% chance it will sunny tomorrow and 50% chance of rainy. CO3 An 5
 1. Draw the state diagram and determine its transition probability matrix.
 2. If today is sunny. What is the chance of rain the day after 3'd day?

3. Now Obtain the steady state probability vector, if it exists, also calculate the percentage of sunny and rainy days after one year.

- 5 b) Electric cars arrive at a charging station that has only one fast-charging point. The arrival of cars follows a Poisson process with an average of 12 cars per hour. The time it takes to fully charge a car is exponentially distributed, with an average charging time of 4 minutes. CO3 An 5

Based on this, calculate the following:

- i) The average waiting time for a car before charging begins.
- ii) The average number of cars waiting in line.
- iii) The probability that an arriving car will have to wait before it can start charging.
- iv) The utilisation factor (traffic intensity) of the charging point.
- v) The probability that there are exactly 3 cars in the system (either charging or waiting).

Bismillahir Rahmanir Raheem
International Islamic University Chittagong
Department of Computer Science and Engineering
B. Sc. in CSE Final Term Examination, Spring 2025
Course Code: CSE 4741
Course Title: Computer Graphics

Time: 2 hours 30 minutes

Total Marks: 50

Answer all the questions. Separate answer scripts must be used for Group A and Group B.

Group A

- | | CO | Marks |
|--|-----|-------|
| 1a) "The rotation transformation will be same for 2D and 3D, if $R_\theta = R_{\theta,K}$ " – Justify your argument. | CO1 | 2 |
| 1b) Use Liang-Barsky Algorithm to clip the lines drawn in Fig-1. | CO3 | 6 |

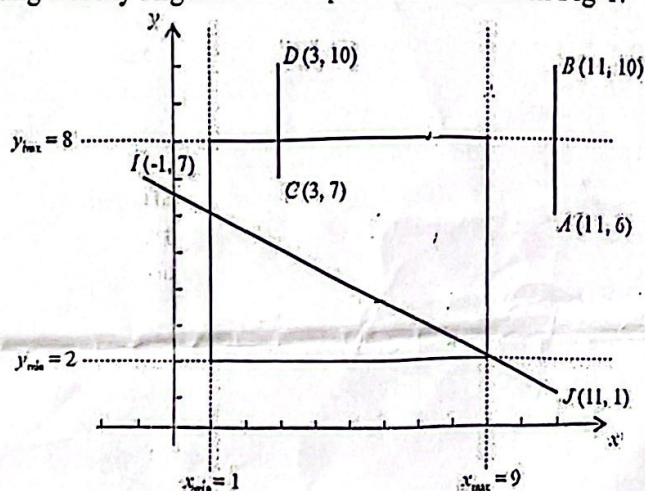


Fig-1: Three lines in 2D plane

OR

- | | | |
|---|-----|---|
| 1b) Use Cohen-Shuterland Algorithm to clip the lines drawn in Fig-1. | CO3 | 6 |
| 1c) Define viewport. Derive the equation for window to viewport mapping. | CO1 | 2 |
| 2a) Differentiate between geometric and coordinate transformation. | CO1 | 2 |
| 2b) Given a 3D object A (1,2,3), B(4,5,6), C(5,6,7) and D(7,8,4). Find the mirror reflection about xy-plane, yz-plane and zx-plane. | CO3 | 4 |
| 2c) Given a line AB in 3D space with coordinate points A (0, 0, 0) and B(3, 2, 3). Apply the rotation on AB, so that the line will completely align with Z axis (keeping the point A fixed). Find the new coordinates after rotation. | CO1 | 4 |

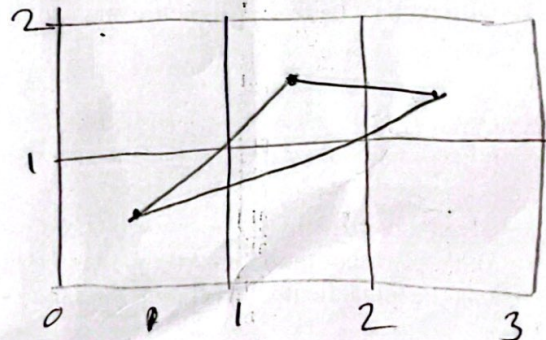
OR

Find the align matrix which aligns a given vector V with the vector K along the positive z-axis.

CO1 4

Group B

- 3a) What is the difference between orthographic and oblique projection. Describe some subcategories of these two projections. CO1 3
- 3b) The unit cube is projected onto the xy plane. Draw the projected image using the standard perspective transformation with
(i) $d = 4$
(ii) $d = 10$
where d is distance from the view plane. CO2 4
- 3c) Explain mathematical models of orthographic and oblique projection. CO1 3
- 4a) Using a 3×2 pixel display, show how the Z-buffer algorithm would determine the color of each pixel for given object A (Triangle) and B (Square) when view-plane is in xy-plane. Where coordinates for object A: (0.5, 0.5, 0), (1.5, 1.5, 0), (2.5, 0.5, 0) and the coordinates for object B: (1.5, 0, 0.5), (3, 0, 0.5), (3, 2, 0.5) and (1.5, 2, 0.5); Assume that, each pixel contain one unit in xy coordinate and z-value is normalized. CO3 5
- 4b) Describe Painter's Algorithm. Why sorting is necessary in this algorithm? CO2 3
- 4c) Justify the method which one is appropriate for shading using interpolation. CO2 2
- 5a) A ray is represented by $r(t) = s + td$, where, $s = 2I + J - 3K$ and $d = I + 2K$. Find the coordinates of the points on the ray that corresponds to $t = 2, 3$ and 4 respectively. CO2 4
- 5b) How ray tracing algorithm can be used to determine shadows of object in 3D representation? Explain with a diagram. CO2 3
- 5c) "Phong model is widely used to mimic the reflection of light from object surfaces to the viewer's eye". Describe the Phong model in brief. CO2 3



International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Spring 2025

Course Code: CSE 4877

Course Title: Machine Learning and Data Mining

Total marks: 50

Time: 2 hours 30 minutes

Group A

CO DL M

1. a) What are the deciding factors to use regression and classification algorithms? Consider k-fold cross-validation. Let's consider the tradeoffs of larger or smaller k (the number of folds). With a higher number of folds, the estimated error will be, on average, (a) Higher (b) Lower (c) Same (d) Can't tell. Explain the reason.

3 U 6

OR

Describe the schemas for Multidimensional Data Models with necessary figures.

1. b) The data tuples are sorted by decreasing probability value, as returned by a classifier. For each tuple, compute the true positive rate (TPR) and false positive rate (FPR). Plot the ROC curve for the data.

4 An 4

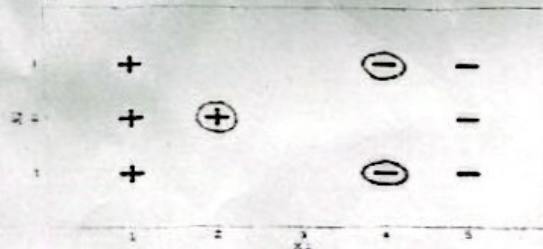
Tuple #	Class	Probability
1	P	0.95
2	N	0.85
3	P	0.78
4	P	0.66
5	P	0.60
6	P	0.55
7	N	0.53
8	N	0.52
9	N	0.51
10	P	0.40

OR

Suppose, you are developing a binary classification model to detect a life-threatening condition from medical images. The dataset is imbalanced, with more "Healthy" (Negative) than "Affected" (Positive) cases. Which metric (accuracy, precision, recall, F1-score) is most appropriate for evaluating the model? Justify considering the impact of false positives and false negatives. Given: TP = 85, TN = 420, FP = 30, FN = 5

2. a) Suppose we are using a linear SVM (i.e., no kernel), with some large C value, and are given the following data set. Draw the decision boundary of linear SVM. Give a brief explanation. Note that, circled points are support vectors.

3 Ap 4



2. b) What is the significance of using ROC curves in machine learning classification?

4 U 2

2. c) Describe Information gain and Gini index attribute selection measure with example.

3 Ap 4

GROUP B

3. a) Describe the Boosting ensemble method to improve the accuracy of a model with necessary figure.

3 U 5

3. b) Explain the K-NN algorithm with appropriate example.

3 U 5

4. Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9) using k-means clustering. Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2). You have to show three epoch.

3 Ap 10

5. a) Give an application example where global outliers, contextual outliers, and collective outliers are all interesting. What are the attributes, and what are the contextual and behavioral attributes? How is the relationship among objects modeled in collective outlier detection?

3 Ap 5

- b) Adapt a simple semi-supervised method for outlier detection. Discuss the scenario where you have (i) only some labeled examples of normal objects, and (ii) only some labeled examples of outliers.

3 Ap 5

OR

Given the points A(3, 7), B(4, 6), C(5, 5), D(6, 4), E(7, 3), F(6, 2), G(7, 2) and H(8, 4), Find the core points and outliers using DBSCAN. Take Eps = 2.5 and MinPts = 3.

3 AP 10